Dollar Stablecoins and Arbitrage across Capital Controls

Evidence from the Kimchi Premium

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Master's Thesis

Outline

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Kimchi premium

- The kimchi premium refers to price discrepancies between Korean and offshore exchanges.
- Marginal investors outside the US and Europe are willing to pay more for bitcoin in response to positive news or sentiment.

$$Premium_{Kimchi} = \frac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1$$

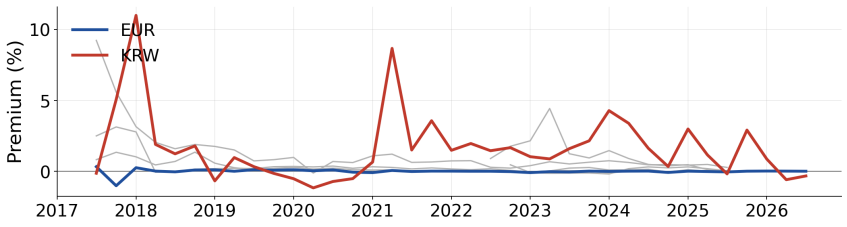
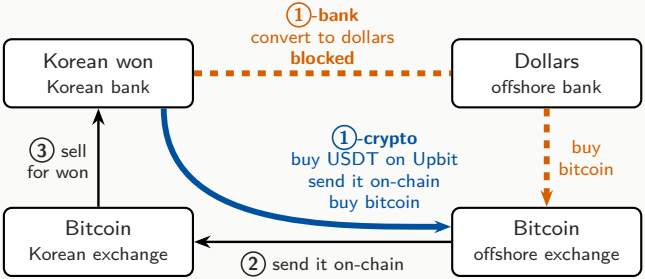


Figure 1: Bitcoin's premium in seven local currencies over Coinbase's USD price

Bitcoin's price on one local exchange per currency over Coinbase's USD price, quarterly mean of daily premia (Makarov & Schoar 2020). Grey: AUD, BRL, GBP, PHP, TRY, to 2025 Q3; KRW and EUR to 2026 Q3.

Two routes for arbitrage: banking system or blockchain network



Makarov & Schoar (2020, *Journal of Financial Economics*)

*"[C]apital controls are most relevant for **the price of crypto to fiat currencies**. (①-**bank**)*
*In contrast, **transactions between two cryptocurrencies** (①-**crypto**) should not be affected [. . .]."*

Paying more for bitcoin, or for dollars?

- **USDT** is a cryptocurrency pegged 1-to-1 to the US dollar that moves through blockchain.

$$\underbrace{Premium_{Kimchi}}_{\text{kimchi premium}} \approx \underbrace{Premium_{Dollar}}_{\substack{\text{fiat-to-crypto spread}\\ \text{Dollar scarcity}}} + \underbrace{Premium_{Crypto}}_{\substack{\text{crypto-to-crypto spread}\\ \text{Local crypto demand}}}$$

$$\frac{P_{\mathrm{KR}}^{BTC/KRW}}{\boldsymbol{FX}^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1 \approx \frac{P_{\mathrm{KR}}^{USDT/KRW}}{\boldsymbol{FX}^{USD/KRW}} - 1 + \frac{P_{\mathrm{KR}}^{BTC/KRW}}{P_{\mathrm{GLB}}^{BTC/USDT} \cdot \boldsymbol{P}_{\mathbf{KR}}^{\boldsymbol{USDT/KRW}}} - 1$$

Research question and Hypotheses

Research Question

Do dollar stablecoins circumvent the capital controls that sustain the kimchi premium?

H1 USDT as an arbitraged target.

- If USDT facilitates cross-border arbitrage, USDT flows arbitrage away the kimchi premium.
- Otherwise, USDT flows narrow USDT/KRW premium over the USD/KRW exchange rate.

H2 USDT as an arbitrage vehicle. USDT flows narrow the crypto premium after removing the foreign-exchange rate component.

Three strands of literature

- **Limits to arbitrage.** Identical assets can trade at different prices across markets because arbitrageurs face capital constraints and risk.
  Shleifer & Vishny (1997, *Journal of Finance*); Froot & Dabora (1999, *Journal of Financial Economics*)

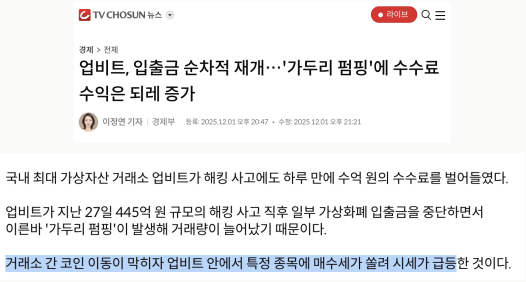
- **The kimchi premium and capital controls.** Crypto premia are largest across borders with capital controls, which bind the fiat-to-crypto price.
  Makarov & Schoar (2020, *Journal of Financial Economics*); Choi, Lehar & Stauffer (2022, working paper, R&R *Journal of Financial and Quantitative Analysis*)

- **Stablecoins in crypto markets.** Stablecoins serve as a common numeraire and an arbitrage vehicle across crypto exchanges. Griffin & Shams (2020, *Journal of Finance*); Kang, Kang, Kim & Sul (2025, *Economics Letters*)

Contribution

1. Decompose the kimchi premium into fiat-to-crypto and crypto-to-crypto components.
2. Exploit crypto transfer suspensions after exchange hacks as quasi-natural experiments.

Fence pumping: market manipulation under the suspension



TV CHOSUN 뉴스 ● 라이브

경제 > 전체

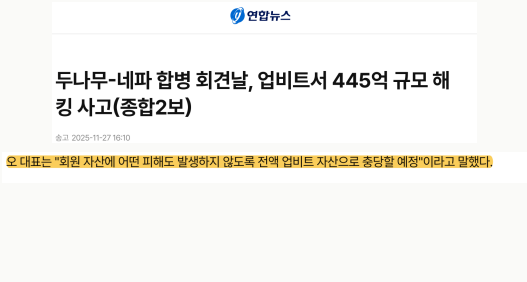
업비트, 입출금 순차적 재개…'가두리 펌핑'에 수수료 수익은 되레 증가

이정연 기자 | 경제부 등록: 2025.12.01 오후 20:47 • 수정: 2025.12.01 오후 21:21

국내 최대 가상자산 거래소 업비트가 해킹 사고에도 하루 만에 수억 원의 수수료를 벌어들였다.

업비트가 지난 27일 445억 원 규모의 해킹 사고 직후 일부 가상화폐 입출금을 중단하면서 이른바 '가두리 펌핑'이 발생해 거래량이 늘어났기 때문이다.

거래소 간 코인 이동이 막히자 업비트 안에서 특정 종목에 매수세가 쏠려 시세가 급등한 것이다.



연합뉴스

두나무-네파 합병 회견날, 업비트서 445억 규모 해킹 사고(종합2보)

송고 2025-11-27 16:10

오 대표는 "회원 자산에 어떤 피해도 발생하지 않도록 전액 업비트 자산으로 충당할 예정"이라고 말했다.

- Upbit **temporarily suspended deposit and withdrawal services** after hacking.
- With suspension, buying pressure concentrated on several cryptos and **pushed their prices up**: 'fence pumping'.

Sources: Yonhap News, 2025-11-27 16:10; TV Chosun, 2025-12-01 20:47. 'Fence pumping': *gaduri pumping*.

Law of one price fails between Upbit and Bithumb during the suspension

- Arbitrage normally keeps one price across exchanges, even within a country.
- Upbit's $Premium_{Kimchi}$ stayed above Bithumb's, which remained open.

(a) $Premium_{Kimchi} = \dfrac{P_{\mathrm{KR}}^{BTC/KRW}}{FX^{USD/KRW} \cdot P_{\mathrm{GLB}}^{BTC/USDT}} - 1$ (b) $Premium_{Kimchi}^{Upbit} - Premium_{Kimchi}^{Bithumb}$

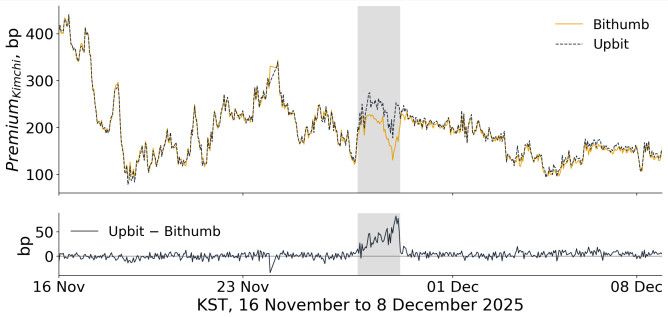


Figure 2: The kimchi premium at Upbit and Bithumb, 16 November to 8 December 2025

Empirical Design: Daily Round-Trip Variables

- j: vehicle cryptoasset
- t: time (daily)

Trading: $NetVol_{x,t} = \sum BuyerInit_{x,t} - \sum SellerInit_{x,t}$

On-chain: $\Delta Rsv_{x,t} = Rsv_{x,t} - Rsv_{x,t+1}$

Round trip arbitrage

$$ArbFlow_{j,t}^{Flow} = \left(Flow_{j,t}\, P_t^{j/BTC} - Flow_{BTC,t}\right)/2$$
$$Flow \in \{NetVol, \Delta Rsv\}$$

Vehicle **Buy** and *withdraw* j crypto from Korea to offshore to buy bitcoin.

Target *Deposit* bitcoin from offshore to Korea to **sell** for KRW.

Followed by Griffin and Shams (2020) to measure the bitcoin arbitrage flow

▸ Sample dataset

Empirical Design: Round Trip and the Kimchi Premium

H1. USDT as an arbitraged target

Daily 2SLS on Upbit's reserves.
the **kimchi premium ($Premium_{Kimchi}$)** or the **USDT/KRW premium ($Premium_{Dollar}$)**?

$$\text{1st stage:} \quad ArbFlow_{j,t}^{NetVol} = \pi_0 + \pi_1\, ArbFlow_{j,t}^{\Delta Rsv} + \pi' W_{t-1} + \pi_2\, 1[FeeDC]_t + u_t$$

$$\text{2nd stage:} \quad \Delta Premium_{Kimchi,t} = \alpha + \beta\, \widehat{ArbFlow_{j,t}^{NetVol}} + \gamma' W_{t-1} + \delta\, 1[FeeDC]_t + \varepsilon_t$$

$$\text{OLS:} \quad \Delta Premium_{Dollar,t} = b_0 + b_1\,(-NetVol_{USDT,t}) + b_2\,(-\Delta Rsv_{USDT,t}) + \lambda' W_{t-1} + b_3\, 1[FeeDC]_t + e_t$$

- Control Variables: $W_{t-1} = \left(\sigma_{t-1},\ \Delta Premium_{Kimchi,t-1},\ \Delta Premium_{Kimchi,t-1}\,\sigma_{t-1}\right)'$
  where σ_t is the standard deviation of $\Delta Premium_{Kimchi}$ over the 20 days to t
- $1[FeeDC]_t$: 1 from Upbit's USDT/KRW fee discount event, 0 otherwise
- OLS: the netflow moves $Premium_{Dollar}$ via Upbit's USDT stock; its W_{t-1} uses $\Delta Premium_{Dollar}$

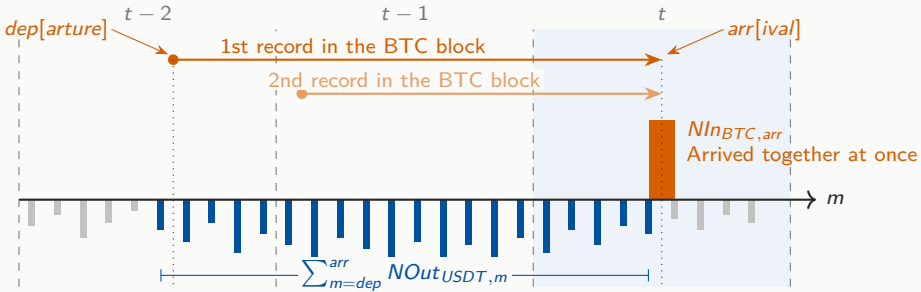
Empirical Design: Convergence of the Crypto Premium

t: 10-minute interval
m: minute
$arr \in (t-1, t]$

Vehicle out: $NOut_{USDT,m} = Out_{USDT,m} - In_{USDT,m}$

Target in: $NIn_{BTC,arr} = In_{BTC,arr} - Out_{BTC,arr}$

$$ArbFlow_{arr} = \left(\sum_{m=dep}^{arr} NOut_{USDT,m} + NIn_{BTC,arr}\right)/2$$



Empirical Design: Convergence of the Crypto Premium

H2. USDT as an arbitrage vehicle

Convergence of $Premium_{Crypto}$ when arbitrage flow arrives

$$\begin{aligned} Premium_{Crypto,t} = \beta_0 &+ \beta_1 \, Premium_{Crypto,t-1} \\ &+ \beta_2 \, ArbFlow_{t-1} + \beta_3 \, Premium_{Crypto,t-1} \times ArbFlow_{t-1} \\ &+ \beta_4 \, Vol_{t-1} + \beta_5 \, Vol_{t-1} \times Premium_{Crypto,t-1} + \varepsilon_t \end{aligned}$$

- $ArbFlow_t = \sum_{arr \in (t-1,t]} ArbFlow_{arr}$
- Arbitrage flow: j leaving, while BTC arrives at Korea (Upbit & Binance/Bybit)
- Vol is the standard deviation of $\Delta Premium_{Crypto}$ over the 24 times intervals before t

Data

12 crypto assets (BTC, ETH, XRP, USDT, . . .) on **Upbit** and **Bithumb**, against **Binance**.

Table 1: Data sources

Data	Frequency	Provider	Coverage	Period
Prices	Hourly, minute	Upbit, Bithumb, Binance, Bybit	12 assets, KRW and USDT	2017-10-24 – 2026-09-25
FX rate	Minute-level	Barchart	USD/KRW interbank	2017-10-24 – 2026-08-28
Trades	Tick-level	Upbit	17 books (BTC, USDT, . . .)	2022-05-01 – 2026-09-25
Holdings	Daily	Arkham Intelligence, XRP Ledger, Hyperion	12 assets, both venues	2017-10-24 – 2026-09-24
Transfers	Transaction-level	Arkham Intelligence, XRP Ledger	6 assets, 2.89M transfers	2019-02-28 – 2026-02-28
Notices		Upbit, Bithumb	501 assets, 2,791 notices	2017-12-12 – 2026-09-24

Data: Summary Statistics

Table 2: Summary statistics: daily throughout, and ten-minute intervals around the 2025 hack

	N	Mean	SD	Min	P5	P50	P95	Max	AR(1)
Panel A: daily, 10 Jun. 2024 ∼ 19 Sep. 2026									
$Premium_{Kimchi,t}$	832	105.8	185.3	-356.6	-159.0	89.3	463.6	858.3	0.93
$NetVol_{BTC,t}$ †	832	-47.16	306.52	-2848.32	-537.15	-27.63	402.28	2172.10	0.22
$\Delta Rsv_{BTC,t}$ †	832	-14.59	281.99	-2009.42	-394.42	-10.87	409.35	1624.85	0.36
$NetVol_{USDT,t}$ ‡	832	11.61	30.24	-110.94	-32.03	7.60	65.81	193.01	0.37
$\Delta Rsv_{USDT,t}$ ‡	832	-0.16	23.74	-158.38	-39.40	1.97	32.03	102.25	-0.08
Panel B: ten-minute intervals, the week before the hack									
$Premium_{Crypto,t}$	983	2.07	6.51	-12.88	-8.67	2.02	12.37	19.79	0.36
$ArbFlow_t$ †	983	1.79	8.20	-38.03	-0.90	0.00	11.94	84.23	0.09
Vol_{t-1}	983	7.40	1.42	4.49	5.14	7.25	10.34	12.06	
Panel C: ten-minute intervals, the week after the reopening									
$Premium_{Crypto,t}$	983	2.21	6.96	-14.67	-10.37	2.46	13.26	16.97	0.39
$ArbFlow_t$ †	983	0.43	2.47	-25.15	0.00	0.00	4.42	17.22	0.04
Vol_{t-1}	983	7.66	1.33	4.31	5.59	7.59	10.09	10.65	

Premia and Vol_{t-1} in bp. † # of BTC. ‡ # of Million USDT.

Data: Correlation

Table 3: Correlation, daily, 10 Jun. 2024 $\sim$ 19 Sep. 2026

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) $\Delta Premium_{Kimchi,t}$	1							
(2) $\Delta Premium_{Dollar,t}$	+0.987	1						
(3) $NetVol_{BTC,t}$	-0.295	-0.295	1					
(4) $\Delta Rsv_{BTC,t}$	-0.069	-0.075	+0.729	1				
(5) $NetVol_{USDT,t}$	+0.511	+0.520	-0.418	-0.195	1			
(6) $\Delta Rsv_{USDT,t}$	+0.417	+0.424	-0.203	-0.078	+0.454	1		
(7) $ArbFlow^{NetVol}_{USDT,t}$	+0.464	+0.469	-0.864	-0.576	+0.808	+0.379	1	
(8) $ArbFlow^{\Delta Rsv}_{USDT,t}$	+0.305	+0.313	-0.636	-0.737	+0.404	+0.718	+0.636	1

Results: Round Trip and the Kimchi Premium

Table 4: USDT flows and Kimchi premium

10 Jun. 2024 ∼ 19 Sep. 2026	Panel A: Kimchi premium USDT as a vehicle		Panel B: Dollar premium USDT as a target
	(1) First stage	(2) Second stage	(3) OLS
$ArbFlow_{j,t}^{\Delta Rsv}$ (π_1)	+0.625*** (+10.63)	–	–
$ArbFlow_{j,t}^{\overline{NetVol}}$ (β)	–	+0.535*** (+9.31)	–
$-NetVol_{USDT,t}$ (b_1)	–	–	-0.527*** (-8.16)
$-\Delta Rsv_{USDT,t}$ (b_2)	–	–	-0.189*** (-4.14)
σ_{t-1}	+0.159*** (+5.92)	-0.127*** (-3.31)	-0.166*** (-4.56)
$\Delta Prem_{t-1}$	-0.010 (-0.10)	-0.493*** (-5.42)	-0.321*** (-3.90)
$\Delta Prem_{t-1}\,\sigma_{t-1}$	+0.200** (+2.08)	+0.341*** (+3.62)	+0.136 (+1.56)
$1[FeeDC]_t$	-0.036 (-0.49)	+0.164* (+1.87)	+0.119 (+1.54)
Constant	+0.008 (+0.27)	-0.036 (-1.07)	-0.026 (-0.84)
First-stage $\chi^2(1)$	112.9		
Endogeneity test, p		0.879	
Observations	832	832	824
Adjusted R^2	+0.463	+0.263	+0.366

Prem: $Premium_{Kimchi}$ in (1), (2) and $Premium_{Dollar}$ in (3).

Results: Convergence of the Crypto Premium

Table 5: Convergence of the crypto premium under arbitrage flow

	Week before the hack			Week after the reopening		
	Full	Neg. lag. premium	Pos. lag. premium	Full	Neg. lag. premium	Pos. lag. premium
Lag $Premium_{Crypto}$	+0.4393**	+0.3034	-0.2655	+0.3892**	+0.7149	+0.0678
	(+2.35)	(+0.67)	(-0.71)	(+2.22)	(+1.23)	(+0.20)
Lag *ArbFlow*	+0.2884	-0.0049	+0.4678**	+0.8623**	+2.6151**	+1.1115
	(+1.62)	(-0.01)	(+2.29)	(+2.04)	(+2.15)	(+1.50)
Lag *ArbFlow* × Lag $Premium_{Crypto}$	+0.0049	-0.0275	-0.0100	-0.1746***	+0.1091	-0.2264**
	(+0.27)	(-0.17)	(-0.48)	(-3.62)	(+0.83)	(-2.47)
Lag *Vol*	+0.2936**	+0.6545*	-0.4232	-0.5285***	-0.5064	-0.8473**
	(+2.03)	(+1.77)	(-1.16)	(-3.33)	(-1.11)	(-2.35)
Lag *Vol* × Lag $Premium_{Crypto}$	-0.0122	+0.0065	+0.0753	-0.0023	-0.0299	+0.0350
	(-0.47)	(+0.10)	(+1.54)	(-0.10)	(-0.42)	(+0.78)
Constant	-0.8256	-3.5716	+4.8325*	+5.4875***	+5.9102	+8.1576***
	(-0.77)	(-1.33)	(+1.76)	(+4.35)	(+1.59)	(+2.99)
Observations	983	372	611	983	370	613
Adjusted R^2	0.128	0.045	0.054	0.171	0.087	0.070

Conclusion and policy implications

Conclusion

1. The kimchi premium sits in the leg exposed to capital control, not in bitcoin, consistent with Makarov and Schoar (2020).
2. Dollar stablecoin does not relax the frictions on the local fiat currency.
3. A dollar stablecoin serves as the vehicle of arbitrage $Premium_{Crypto}$.

Policy implications

1. Listing a dollar stablecoin in a local fiat market does not ensure that its price tracks the exchange rate.
2. In Korea, dollar stablecoins traded at a premium to the U.S. dollar.
3. Capital controls force arbitrageurs to hold bitcoin in Korea in advance, exposing them to price risk.

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Appendix

Korea won's capital controls

Table A1: Korea's capital controls

Rule	In force	Source
Remittance cap	A resident may send abroad without documents at most **USD 50,000** a year (1999 to 3 July 2023), **USD 100,000** a year since 4 July 2023, and **USD 5,000** per transfer (from Jan 2026)	Ministry of Economy and Finance (10 Feb 2023; 8 Dec 2025)
Exchange access	Won deposits require a real-name bank account; **non-residents cannot trade** on Korean exchanges (since 30 Jan 2018)	Financial Services Commission (23 Jan 2018)
Enforcement	**USD 12.26 billion** of outward remittances by 84 firms, collected from crypto exchanges and sent abroad as import payments, 2021–22	Financial Supervisory Service (4 Apr 2023)

Suspend the route, and the pressure stays in the crypto premium

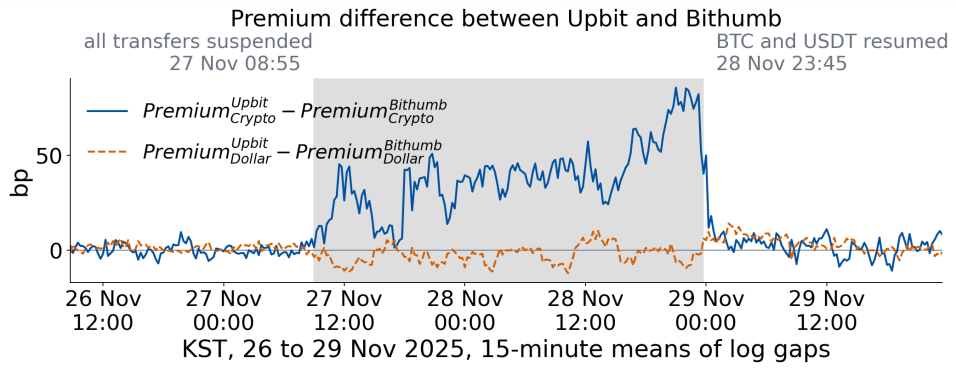


Figure A1: Upbit less Bithumb in the dollar and crypto premia around the suspension

- With suspension, Upbit sellers could sell only what they already held.
- As a result, $Premium_{Crypto}$ spikes while $Premium_{Dollar}$ stays flat.

Daily Round-Trip Variables: Example Data

Trading: *NetVol*

$$NetVol_{x,t} = \sum BuyerInit_{x,t} - \sum SellerInit_{x,t}$$

Timestamp	price (KRW)	volume (BTC)	side
... 12:00:01.659	123,435,000	*0.000041*	*buy*
... 12:00:01.843	123,435,000	*0.000041*	*buy*
... 12:00:01.852	123,427,000	0.003403	sell
... 12:00:01.943	123,427,000	0.000084	sell
... 12:00:02.037	123,435,000	*0.000041*	*buy*
	⋮		

$\sum BuyerInit_{BTC}$, every *buy* that day	826.98
$\sum SellerInit_{BTC}$, every *sell* that day	799.23
$NetVol_{BTC}$	+27.75
$NetVol_{USDT}$, the same from KRW-USDT (M)	+10.23

the regressor, instrumented

On-chain: ΔRsv

$$\Delta Rsv_{x,t} = Rsv_{x,t} - Rsv_{x,t+1}$$

	BTC		USDT (M)	
day t	Rsv_t	ΔRsv_t	Rsv_t	ΔRsv_t
14 Apr	175,645.72	+16.03	126.20	−4.18
15 Apr	175,629.68	+90.08	130.38	+7.26
16 Apr	175,539.60	−115.37	123.12	+1.24
17 Apr	175,654.98	−33.97	121.88	−5.27
18 Apr	175,688.95	−13.97	127.15	−4.29
19 Apr	175,702.92		131.44	

the instrument

◂ Round-Trip Variables

Table A2: Variable definitions

Variable	Definition
$Premium_{Kimchi,t}$	$P_t^{BTC/KRW} \,/\, \left(FX_t^{USD/KRW} \cdot P_{GLB,t}^{BTC/USDT}\right) - 1$
$Premium_{Dollar,t}$	$P_t^{USDT/KRW} \,/\, FX_t^{USD/KRW} - 1$
$Premium_{Crypto,j,t}$	$P_t^{BTC/KRW} \,/\, \left(P_{GLB,t}^{BTC/USDT} \cdot P_t^{j/KRW} / P_{GLB,t}^{j/USDT}\right) - 1$ s.t. j = USDT on H2
$\Delta Rsv_{x,t}$	$Rsv_{x,t} - Rsv_{x,t+1} = Out_{x,t} - In_{x,t}$ s.t. Upbit
$NetVol_{x,t}$	$\sum BuyerInit_{x,t} - \sum SellerInit_{x,t}$
$ArbFlow_{j,t}^{Flow}$	$\left(Flow_{j,t}\, P_t^{j/BTC} - Flow_{BTC,t}\right)/2,\;\; P_t^{j/BTC} = P_{Binance,t}^{j/USDT} / P_{Binance,t}^{BTC/USDT}$ s.t. $Flow \in \{NetVol, \Delta Rsv\}$
$ArbFlow_{arr}$	$\left(\sum_{m=dep}^{arr} NOut_{USDT,m} + NIn_{BTC,arr}\right)/2, \quad ArbFlow_t = \sum_{arr \in t} ArbFlow_{arr}$ s.t. t: ten-minute interval
$NOut_{USDT,m}$, $NIn_{BTC,m}$	$Out_{USDT,m} - In_{USDT,m}, \quad In_{BTC,m} - Out_{BTC,m}$ s.t. m: minute; dep, arr: a BTC transfer's departure and arrival
Vol_{t-1}	standard deviation of $\Delta Premium_{Crypto}$ over the 24 ten-minute intervals before t
$Notice_t$	1 from each Upbit notice stopping or delaying USDT withdrawals to six hours after its reopening

The thesis's tables write K, D and C; the slides write $Premium_{Kimchi}$, $Premium_{Dollar}$ and $Premium_{Crypto}$.